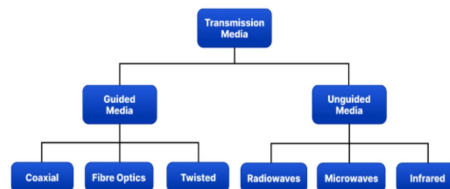


Transmission Media

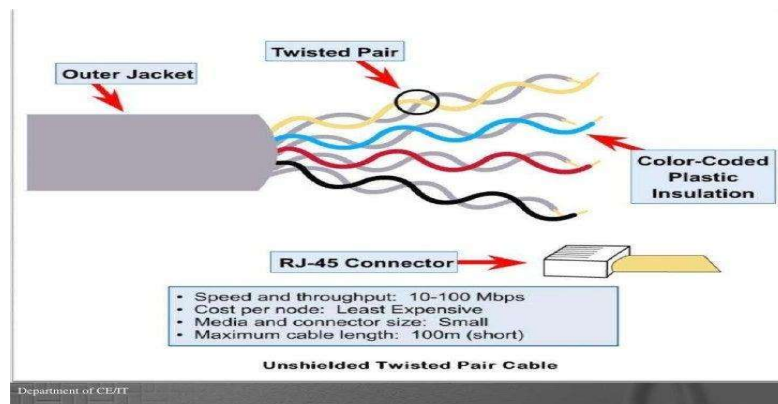
Transmission media refers to the **physical or wireless path through which data signals travel from a sender**



Wired Communication Media

Wired media use a **physical cable** to transmit data signals.

A. Twisted-Pair Cable



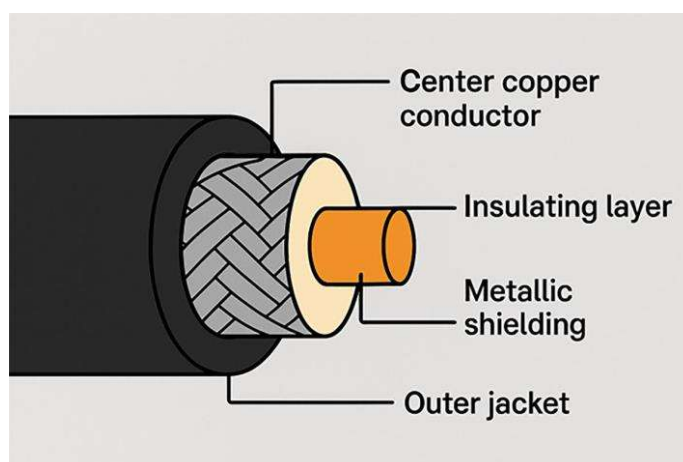
Key Points

- Consists of **two insulated copper wires twisted around each other**.
- Twisting helps reduce **electromagnetic interference and crosstalk**.
- It is relatively **inexpensive and easy to install**.
- Commonly used in **telephone lines and computer networks**.
- Two common types:
 - **UTP – Unshielded Twisted Pair**
 - **STP – Shielded Twisted Pair**
- Usually connected to network devices using an **RJ45 connector**.

Advantages: Low cost, easy installation

Limitation: More susceptible to interference and has shorter range compared with fiber optic.

B . Coaxial Cable



A coaxial cable consists of a **central conductor** surrounded by insulation, metallic shielding, and an outer protective covering.

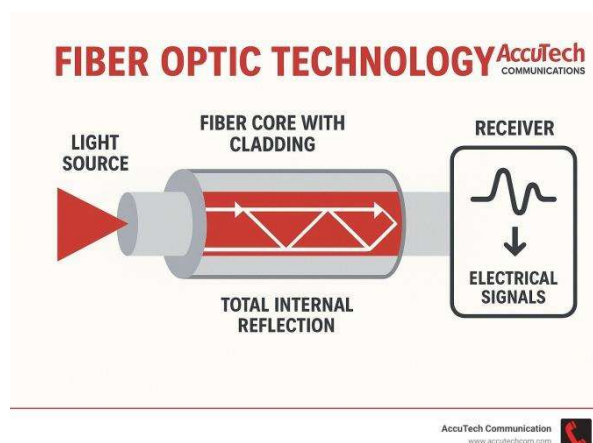
Key Points

- Has a **central copper conductor**.
- Surrounded by an **insulating layer**.
- A metallic shield protects the signal from interference.
- Outer jacket provides physical protection.
- Has better shielding against interference than twisted pair.
- Used in **cable television, broadband connections, and older computer networks**.

Advantages: Better protection against noise; relatively high bandwidth

Limitation: Thicker and less flexible than twisted pair.

C. Fiber-Optic Cable



Fiber-optic cable transmits data in the form of **light signals** through very thin strands of glass or plastic.

Key Points

- Uses **light signals** instead of electrical signals.
- Made mainly of **glass or plastic fibers**.
- **Core** carries the light signal.

- **Cladding** surrounds the core and helps keep light within it.
- Provides **very high bandwidth and high-speed communication**.
- Suitable for **long-distance communication**.
- Not affected by electromagnetic interference.
- More expensive and comparatively difficult to install than copper cables.

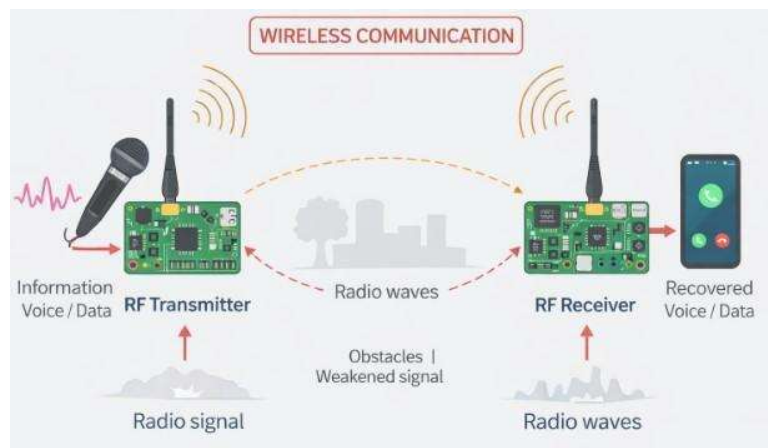
Advantages: Very high speed, high bandwidth, long distance, low interference

Limitation: Higher installation cost and requires specialized handling.

Wireless Communication Media

Radio Waves

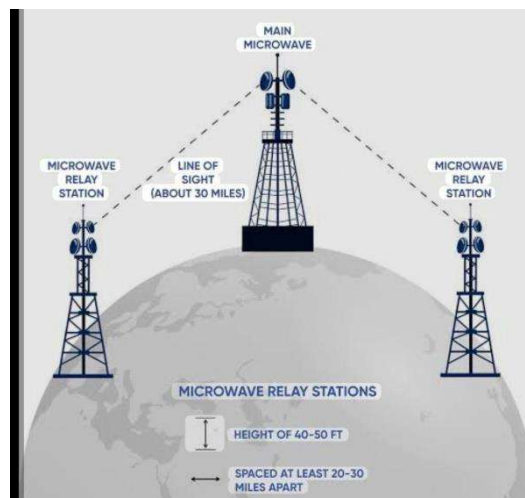
Wireless media transmit data **without a physical cable**, generally using electromagnetic waves.



Key Points

- Use **radio-frequency electromagnetic waves**.
- Generally suitable for **wireless communication over relatively large areas**.
- Can travel through air and, depending on frequency and conditions, may pass through some obstacles.
- Used in **radio broadcasting, wireless communication, and some computer networking applications**.
- Can support communication without requiring a direct physical cable.

Microwaves



Key Points

- Microwaves are **high-frequency electromagnetic waves**.
- Commonly used for **wireless communication over long distances**.
- Terrestrial microwave communication generally requires **line-of-sight** between transmitting and receiving antennas.
- Used in **telecommunication links, satellite communication, and wireless networks**.
- Suitable for carrying large amounts of information.

Infrared Waves

Key Points

- Infrared uses **infrared electromagnetic radiation**.
- Generally used for **short-range communication**.
- Usually requires a relatively direct path between transmitter and receiver.
- Cannot normally penetrate walls.
- Used in **remote controls and short-range device communication**.
- Less suitable for long-distance networking.